Stabilization Programs and the Consumption of Durable Goods

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Abstract

This paper examines the consumption boom that follows price stabilization programs based on price or exchange rate freezes, with a focus on durable goods consumption — a dimension typically neglected in the literature. We develop a model in which households are heterogeneous in income and in their access to credit markets, and must save to purchase a durable good, exposing themselves to the erosive effects of inflation. The effective price of the durable good depends intrinsically on the inflation rate. We use the model to simulate the effects of the Cruzado Plan — Brazil's main heterodox stabilization episode — on refrigerator ownership. Results show that a permanent stabilization generates an expansion of approximately 5 percentage points in durable goods ownership over one year, while a temporary ten-month stabilization generates an expansion of 1.2 percentage points. Incorporating preference heterogeneity improves the model's fit to the data.

1 Introduction

During the 1980s, several countries suffered from episodes of high inflation: Argentina, Chile, Uruguay, Israel, and Brazil, for example. But the similarity among these countries goes beyond the inflationary situation. At some point during the 1980s, all of these countries bet on plans whose focus was on price or exchange rate freezes. The objective of these plans was simple: to defeat the inflationary problem that seemed beyond the control of more orthodox stabilization plans.

Interestingly, although in the long run most of these plans failed, in the short run they not only reduced inflation but also managed to combine this reduction with a considerable expansion of consumption. De Gregorio et al. (1998) list some of these events. According to the authors, in the first year after its Convertibility Plan (1991), Argentina experienced an expansion of 15.7% in total consumption (private plus public), reaching a maximum expansion of 37.6% in total consumption before a recession. The expansion in car sales was much larger: 400% in the same period. Israel, in turn, experienced an expansion of approximately 25% in non-durable goods consumption between 1985 and 1988, while durable goods consumption increased by 70%. In Chile, over a 3-year interval after the plan's implementation (1978–81), durable goods consumption more than doubled, while total consumption grew 26%. For Brazil, in the year of the Cruzado Plan, while the National Consumer Price Index (INPC) went from something between 13% and 15% in the months before the freeze to 3.18% in March and less than 1.5% in the following five months, durable goods production grew 20.4% and non-durable goods production grew 9.0%.[1]

From these experiences, two facts seem clear. First, such sudden reductions in inflation are typically followed by expansion cycles. Second, these cycles are generally larger for durable goods than for non-durables.

This paper addresses these two aspects of the post-stabilization period. The proposed model has a framework in which households differ, first, by their endowment, and second, by their access or not to the credit market after purchasing the durable good. Households must choose between saving or not in order to purchase a durable good, the opportunity cost being the non-durable goods they will forgo consuming. The main point is that the effective price of the durable good depends inextricably on the inflation rate.

In this framework, the expansion arises from the reduction in the effective price of the durable good that occurs with price stabilization. If the reduction in inflation is sufficiently large, the change in the effective price will cause households that previously did not wish to purchase the durable good to start saving for that purpose, generating the expansion.

The model is interesting because it allows not only the study of a permanent price

[1] Data source: IBGE.

stabilization, but also of a temporary stabilization. In addition, it enables the study of the effects of a reduction in the inflation rate on those households that, with high inflation, would not purchase the good (the extensive margin) and also on those that would already purchase it regardless of whether inflation was at the pre- or post-stabilization level (the intensive margin). Another advantage is the wide range of model variations possible, including assigning a preference parameter to each household in the economy.

First, a framework is presented in which households are unable to access the credit market, requiring them to save the full value of the durable good before purchasing it, thus bearing the full effects of inflation. Subsequently, the possibility is introduced of households paying only a down payment for the durable good and financing the remainder through a loan. The purpose of this model is to make the credit financing possibility more realistic.

To evaluate the model's implications, it is used to simulate the effects of the Cruzado Plan — the main price-freeze stabilization episode in Brazil — on refrigerator ownership.[2] Results show that the magnitude of the expansion grows with the duration of the stabilization: 5 percentage points for a permanent stabilization, 1.2 for a ten-month stabilization.

In addition, several robustness checks are performed to assess the various effects the model has on ownership of durable and non-durable goods.

Finally, the model fits the data better when we include the possibility of preference heterogeneity in the model with a transitory shock.

It is worth noting that the expansion in the simulation arises from the fourth quartile, which has the largest expansion in ownership according to the National Household Sample Survey (PNAD). Thus, the model focuses on the correct households. Furthermore, the reduction in inflation, holding other parameters constant, leads households to save for longer periods in the simulation, since it reduces the cost of consumption smoothing.

2 Literature Review

The literature has presented some models to address the relationship between this specific type of price stabilization and the consumption boom. Calvo (1986) frames the expansion as a direct result of price controls through his *temporariness hypothesis*. If individuals believe the program is likely to fail, then the temporary reduction in inflation implies a reduction in the nominal interest rate, which under a cash-in-advance constraint also translates into a fall in the effective price of consumption during the freeze period. This

[2]The refrigerator item is used because it is one of the two available in the PNADs of that period. The second item, the stove, was already present in the vast majority of households in the 1980s, and most of the effect of the end of inflation on its consumption should occur through the replacement of a depreciated good, which is not captured by the PNAD.

change in the intertemporal relative price of consumption leads to an expansion during the period of price controls and a permanent reduction after the return of high inflation.

Although one of the most accepted explanations, Calvo's proposal was subject to criticism. First, while his argument remains valid as long as there is a positive probability of plan failure, more credible stabilization plans should produce smaller expansions (if any at all), which does not necessarily occur, as Bruno (1993) argues. Moreover, since estimated intertemporal elasticities tend to be very low, there are doubts about the quantitative importance of the *temporariness hypothesis* (De Gregorio et al., 1998).

Rebelo (1997) posits that an exchange rate freeze policy must be accompanied by a fiscal reform to succeed. Thus, if the reform occurs earlier than individuals expect, there is a positive shock to expected income and consumption grows. Helpman and Razin (1987) construct a model in which a price freeze without a reduction in the government deficit leads to an intertemporal redistribution of income and, therefore, to an expansion.

A limitation runs through both articles. Both seek the income effect in government action, not in the reduction of inflation per se. In fact, they start from opposing hypotheses: while Rebelo (1997)'s argument depends on the connection between exchange rate control and fiscal reform, Helpman and Razin (1987) assume exactly the opposite — that no fiscal reform accompanies the freeze.

Furthermore, Calvo (1986), Helpman and Razin (1987), and Rebelo (1997) provide no internal reason within the model to explain the larger expansion in durable goods. All three papers ignore the role of these goods during the expansion.

The only paper to focus on durable consumer goods is De Gregorio et al. (1998). Their objective, however, is not to explain the origin of the boom. The authors' argument is simpler and aims to explain why the expansion arising from the positive income impact of reduced inflation is generally followed by a recession a few years later. To do so, they use an (S,s) model of durable goods consumption and show that the recession is endogenous to the original expansionary process itself: the income shock leads agents to advance their purchases (boom), and since durable goods are purchased discretely, the very act of agents advancing their purchases leads to a future recession, as none of them will wish to purchase until the good has depreciated sufficiently.

The paper, however, does not address the possibility of temporary shocks. In the model's resolution, stabilization is assumed to be permanent. Thus, the model presented by De Gregorio et al. (1998) appears less flexible than the one presented here.

Therefore, this paper differs from the others: first, by focusing on durable goods consumption, as does De Gregorio et al. (1998); unlike that work, however, it evaluates both permanent and temporary price level stabilizations. It is also noteworthy that the effect of inflation on the intensive margin differs between the two papers. While in De Gregorio et al. (1998) a reduction in inflation leads to a frontloading of consumption, in the simulations presented here a reduction in inflation will cause households to smooth consumption

for longer, increasing the average saving time.

The paper also differs from Helpman and Razin (1987) and Rebelo (1997) in that the expansion does not depend on events beyond the simple stabilization of inflation. Additionally, in contrast to Calvo (1986), the higher the credibility and duration of the plan, the larger the expansion in consumption.

3 Historical Overview

This section presents in detail the historical period of 1985–87, during which Brazil underwent two distinct heterodox inflation control programs — the Cruzado Plan.

3.1 The Cruzado Plan

The stabilization plan that frames the period of interest for this paper was born in a turbulent moment of national politics, which would tremendously influence not only its design but also its execution. After a long period of military and dictatorial governments, the "Diretas Já" movement and the dissatisfaction with the political model among part of the political and military establishment foreshadowed the change that would come with the first (indirect) election of a civilian president in 20 years, Tancredo Neves. However, due to health problems, the president-elect died before taking office, and his vice-president, José Sarney, assumed the role. As an unelected president, Sarney took office without electoral legitimacy, which led him to guide his government in an attempt to earn it in the streets (Castro, 2005). The Cruzado Plan was marked by this fact.

In 1985, the economy was booming — GDP growth was 7.85% for the year and durable goods production grew by 14.51% — and external accounts ended the year in surplus. After a year of conventional measures aimed at combating inflation, however, the INPC reached 239% per year, making it clear to the government that more drastic actions were needed to decelerate the inflationary process. Based on the assessment that Brazilian inflation had an inertial character, the then Finance Minister Dílson Funaro implemented the Cruzado Plan on February 28, 1986, which would be the first in a series of heterodox plans.

A new currency was created, the Cruzado, which replaced the Cruzeiro at a rate of 1,000 to 1. Prices were frozen at their prevailing values, and wages were fixed based on their average purchasing power over the previous six months. Meeting popular demands for income redistribution, an 8% bonus was granted to all salaried workers and the minimum wage was increased by 16%. Annual collective wage agreements were reinstated, under which wages were to be automatically adjusted by at least 60% of the change in the cost of living over the period. In addition, a wage trigger was established, whereby wages would be automatically adjusted by 20% once accumulated inflation reached that

figure (Modiano, 1990).

The exchange rate was also fixed at its February 27 value and none of the taxes were adjusted, with the exception of industrial electricity tariffs, which were increased by 20%. Indexation of contracts with duration shorter than one year was prohibited, and savings accounts moved to quarterly rather than monthly adjustments. The ORTN,[3] corrected monthly on the basis of the official inflation index, was replaced by the OTN, whose nominal value was fixed for 12 months. Finally, no targets were set for fiscal or monetary policy (Modiano, 1990).

Modiano (1990) divides the analysis of the Cruzado Plan results into three periods. The first, covering the months from March to June 1986, was characterized by a sharp fall in inflation, whose peak in March was 3.18% per month according to the INPC. Figure 1 shows the monthly inflation for the years 1985 to 1987. The initial success earned the plan enormous popular support, with the population even taking on the task of monitoring prices — giving rise to the so-called "Sarney price inspectors."

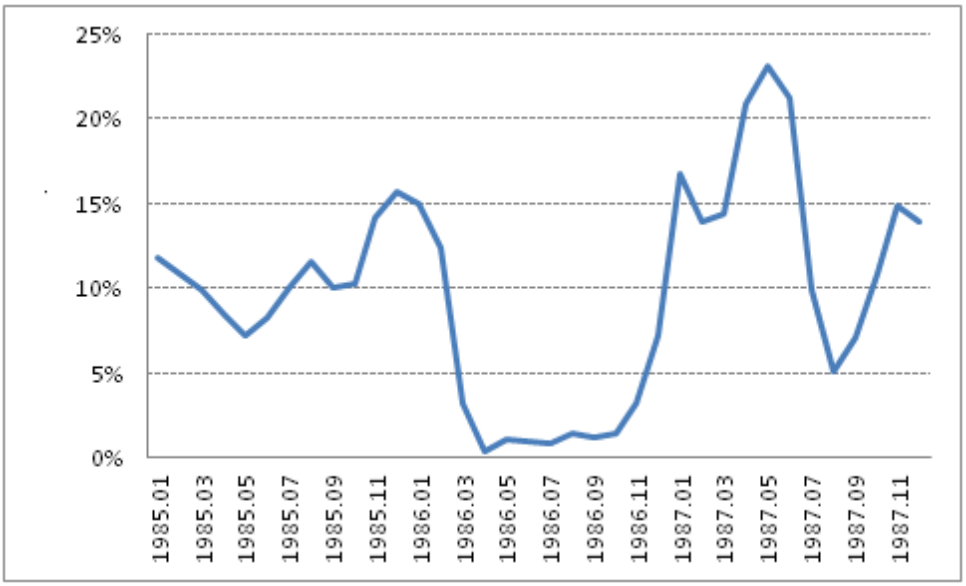


Figure 1: Monthly INPC variation, 1985–87.
Source: Own elaboration. Data: IBGE.

Some indicators, however, already signaled overheating demand: prices that were difficult to control, such as clothing and used cars (which together account for 15% of the official index), consistently increased by 4% to 5% per month. Modiano attributes this demand overheating to "the increase in purchasing power of wages, the voluntary dissaving caused by monetary illusion, the decline in personal income tax collection, the reduction in nominal interest rates, consumption pent up during the recession years, and the freezing of some prices at levels lagging behind their costs" (Modiano, 1990, p. 361).

[3]Obrigação Reajustável do Tesouro Nacional (Readjustable National Treasury Bond), a Treasury-issued bond adjusted monthly according to the official inflation index.

Also during this period, not only did the government maintain a monetary policy contradictory to the inflation control objective — political pressures led to the maintenance of a low nominal interest rate and an excessively large expansion in money supply — but the problems of public deficit caused by tax reductions, increased transfers, subsidies, and wages also became apparent.

In the second period, from July to October 1986, the government unsuccessfully tried to cool consumption with a set of fiscal measures known as the Cruzadinho. Price changes caused by government policies were excluded from the CPI (IPC) in order to prevent the wage trigger from being activated. Although official inflation remained under control, the premium (ágio) spread through the economy.

The final period covers November 1986 to June 1987 and is marked by the recognition of the Cruzado Plan's failure. One week after the elections for governor and for the Constituent Assembly, the Sarney government launched Cruzado II, a new set of fiscal reforms with tax increases, whose impact the government tried to exclude from the CPI, thus preventing the activation of the wage trigger. Due to popular pressure, the government reversed course on the exclusion, causing an automatic 20% increase in wages. Fearing the imminent return of inflation, the government began a process of re-indexing the economy, completed in February, when the price freeze was lifted and the debt moratorium was declared.

4 Data

This section of the paper discusses the data sources used both for the future calibration of parameters and for the analysis of the period covering 1985, 1986, and 1987. In addition to presenting these data, a brief exploratory analysis is conducted with two objectives: to better understand the durable goods market at that time and to argue why the model's basic assumption — the absence of a credit market — makes sense for the Brazilian economy of the 1980s. It is worth highlighting the difficulty of obtaining certain information relating to this period, in particular the price of consumer durable goods. For this reason, a database was constructed from newspaper advertisements of the time, as will be explained below.

4.1 Data Sources

The PNAD is an annual survey conducted by IBGE using September as its reference month. This paper uses data from the PNADs of 1984 to 1988, particularly data on durable goods and income.[4]

[4]Any PNAD household entries for which income, refrigerator ownership, or another control variable were undetermined were excluded. For the calculation of average income, the wealthiest 1% was also excluded to avoid the influence of potential outliers.

The durable goods questions ask whether the household owns a stove and whether the household owns a refrigerator. In neither case is the quantity or the date of acquisition specified; thus, PNAD data ignore any stock replacement effects. Since the stove is an older and cheaper good than most durables, the percentage of the population that owns one is very high across all income levels. Thus, any income expansion effect occurs through the replacement of existing goods rather than through changes in ownership, which does not show up in PNAD data. For this reason, this paper uses only data on refrigerator ownership. In addition to this variable, monthly household income will also be used in constructing the parameters to calibrate the model.

As a proxy for the price of the non-durable goods basket, the National Basic Basket Survey (Pesquisa Nacional da Cesta Básica) was used, administered by the Inter-Union Department of Statistics and Socioeconomic Studies (DIEESE). This survey provides the current price of a basic basket composed of the following items: meat, milk, beans, rice, flour, potatoes, tomatoes, bread, coffee, bananas, sugar, oil, and butter. In 1985, this data is available for seven cities. For reasons discussed in the calibration section, only São Paulo data were used.

No database was found with a price series for durable goods. The alternative was to construct a database from newspaper advertisements from the 1980s. All advertisements for televisions, refrigerators, and stoves appearing in Sunday editions of the newspaper "O Estado de São Paulo" between January 1985 and December 1987 were catalogued. Although the simulation is performed only for refrigerators, data on stoves and televisions were collected to provide a better understanding of financing arrangements. This research was conducted using the collection of the Arquivo do Estado de São Paulo, maintained by the state government. The database entries are as follows: date, store name, type of good, model, cash price, number of installments, installment value, and whether a down payment is required.[5] In addition, the following were calculated: the inflation-adjusted price of the good (base September 1985), the nominal interest rate in the purchase period (dividing the installment price by the cash price), the real interest rate in the purchase period, and the monthly nominal and real interest rates. Whenever necessary, the INPC was used to standardize the series at September 1985 values.

Finally, although not used in the simulations, the 1987/88 POF (Family Budget Survey) was used to provide additional data on durable goods purchases. The POF is a sample survey conducted by IBGE with an approximate periodicity of seven years (with no commitment to maintaining this periodicity), aimed at producing information on both family and individual income and expenditure. It includes data on durable goods spending and, most importantly, on how these goods were acquired: cash, installments, etc.

[5]One could argue that companies advertise only what they consider their best offers. In that case, this database would be biased downward in both goods prices and in implicit interest rates.

4.2 Analysis

Table 1 shows the percentage of each PNAD income quintile that owns a refrigerator, by year. As can be seen, this percentage is highly sensitive to income. While in the fifth quintile the share of households is below 30%, in the first quintile this figure reaches almost 100%.

Table 1: Percentage of population with a refrigerator, by income quintile

Year	Total	5th Quintile	4th Quintile	3rd Quintile	2nd Quintile	1st Quintile
1984	62.7%	22.0%	43.3%	66.6%	84.9%	96.7%
1985	63.9%	22.6%	44.6%	68.0%	86.5%	96.8%
1986	65.7%	25.5%	49.5%	70.9%	86.3%	96.6%
1987	66.7%	25.1%	50.1%	72.8%	88.3%	97.5%
1988	68.0%	27.5%	52.2%	73.9%	89.0%	97.4%

Source: Own calculations. Data: PNAD.

There is a more pronounced increase in this percentage in 1986, particularly for the fourth and fifth quintiles: approximately 3 pp and 5 pp, respectively. These figures point to a larger expansion in the year of the Cruzado Plan, but could also be explained by other factors not being controlled for, such as wage bonuses. For this reason, a Linear Probability Model (LPM) was estimated by regressing refrigerator ownership on year dummies, real household income, and other relevant controls. The estimated equation was:

$$\text{fridge}_i = ad85_i + ad86_i + ad87_i + ad88_i + \text{controls}_i$$

where fridge_i is the dependent variable, equal to 1 if the household owns a refrigerator at the time of the interview; $ad85_i$ is a year dummy equal to 1 if the household belongs to the 1985 PNAD sample or later; $ad86_i$ is a year dummy equal to 1 if the household belongs to the 1986 PNAD sample or later, and analogously for $ad87_i$ and $ad88_i$; controls_i are the controls for that household.[6].

Table 2 below shows the values found for the year dummies. It is important to recall that, as constructed, the correct interpretation of the year dummies is the marginal effect of that specific year on the probability of a household owning a refrigerator. The coefficient of $ad87$ being 0.76 for the second quintile means that a second-quintile household in the 1987 PNAD has a 0.76 pp higher chance of owning a refrigerator than a household from the same quintile in the 1986 sample.

[6]The list of controls is in Appendix .

Table 2: Linear Probability Model

Variable	Total	5th Quintile	4th Quintile	3rd Quintile	2nd Quintile	1st Quintile
$ad85$	-0.0159^{***}	-0.00461	-0.000303	-0.00639	-0.00867^{***}	-0.00342^{**}
	(0.00161)	(0.00373)	(0.00435)	(0.00394)	(0.00290)	(0.00140)
$ad86$	-0.000525	0.00795^{*}	0.0157^{***}	0.00215	-0.0110^{***}	-0.00314^{*}
	(0.00179)	(0.00417)	(0.00476)	(0.00422)	(0.00328)	(0.00167)
$ad87$	0.000879	0.00336	0.00731	0.0051	0.00761^{**}	0.00226
	(0.00191)	(0.00447)	(0.00505)	(0.00445)	(0.00342)	(0.00172)
$ad88$	0.00530^{***}	0.0207^{***}	0.0203^{***}	0.0167^{***}	0.00947^{***}	-0.00094
	(0.00189)	(0.00441)	(0.00506)	(0.00451)	(0.00327)	(0.00160)

* significant at 10%; ** significant at 5%; *** significant at 1%.

Source: Own calculations. Data: PNAD.

The regression with the full sample indicates a negligible and insignificant effect for the years 1986 and 1987, both freeze years. More interesting results are obtained for the five separate quintiles. For the three wealthier quintiles, these years show a barely significant or even negative effect on refrigerator ownership; this result is likely related to the large percentage of households that already own one. For these classes, the effect of the end of inflation on durable goods consumption generally occurs through the replacement of an already depreciated good — a household acquires a newer refrigerator, for instance.

This is not the case for the fourth and fifth quintiles. The effect is positive and significant in 1986 and non-significant in 1987. For the fourth quintile, the freeze year represents an increase of more than 1.5 percentage points in the probability of owning a refrigerator, while for the fifth quintile this effect is 0.8 percentage points. These parameters are consistent with an expansion-and-contraction cycle: expansion in 1986 as more durable goods are acquired thanks to reduced inflation, and a contraction in the following period, since most of the households that were about to acquire a refrigerator took the opportunity to do so during the Cruzado Plan.

Another relevant piece of information is that the 1986 dummy effect was almost double for the fourth quintile relative to the fifth. A possible explanation is that in the fourth quintile, there is a much higher share of households that only marginally prefer not to acquire the durable good during high inflation. When there is a significant reduction in inflation, these are the households that will start purchasing the durable good and cause the dummy effect to be larger in the fourth than in the fifth quintile.

Data on how durable goods are acquired are available in the 1987/1988 POF. Table 3 breaks down durable goods purchases by income quintile into cash and installment purchases. The share of refrigerators purchased with cash is relatively high, around 30% for the four lower quintiles and around 50% for the wealthiest quintile.

Table 3: Cash and installment purchases by income quintile

	1984	1985	1986	1987
Fifth Quintile				
Cash purchases	30.3%	33.8%	28.1%	30.8%
Installment purchases	69.70%	66.20%	71.90%	69.20%
Observations	297	320	385	169
Fourth Quintile				
Cash purchases	28.80%	28.10%	26.00%	33.20%
Installment purchases	71.20%	71.90%	74.00%	66.80%
Observations	438	480	596	337
Third Quintile				
Cash purchases	29.50%	26.70%	27.80%	34.30%
Installment purchases	70.50%	73.30%	72.20%	65.70%
Observations	580	643	814	417
Second Quintile				
Cash purchases	31.00%	32.10%	34.10%	42.70%
Installment purchases	69.00%	67.90%	65.90%	57.30%
Observations	715	851	984	501
First Quintile				
Cash purchases	49.70%	52.50%	52.20%	62.80%
Installment purchases	50.30%	47.50%	47.80%	37.20%
Observations	1058	1312	1482	829

Source: Own calculations. Data: POF.

These installment purchases can be further explored using the newspaper advertisement data. From this source, two clearly distinct forms of installment purchase are apparent. The first requires a down payment at the time of purchase and the number of additional installments does not exceed 3, frequently being 2 extra payments. In this case, the interest rate charged by the store is minimal, if not zero.

The argument made here is that this mode of purchase, although on installments, is not so different from what is presented in the model. From the advertisement database and the PNAD, it is possible to calculate the proportion that the price of a refrigerator represents of the median household income. To do so, the median inflation-adjusted refrigerator price was divided by the median household income in 1985, yielding a value of 142%. This means that if households had no prior savings, they would have to commit more than 30% of their total income each month to installments, which is an unlikely alternative under the consumption smoothing hypothesis. But if they have to save a share of the durable good's price before the purchase, then the situation is similar to that presented in the model, with the caveat that the household would only need to save a (high) percentage of the durable's price rather than its full price.

The case of purchases with longer payment periods is somewhat more complex. These

are typically made without a down payment and with a positive real interest rate. The terms extend up to twelve installments, and the real monthly interest rate varies across different periods. Between early 1985 and the beginning of the Cruzado Plan, it was around 9% per month (with a peak at Christmas with monthly rates of up to 35%), fell to approximately 1.5% to 2.5% during the freeze, rose to approximately 6.5% after its failure, and fell again during the three months of the Bresser Plan.

In short, purchasing the durable good before the stabilization plan, an individual would pay a real monthly interest rate of 10%, while during the Cruzado Plan the rate was no more than 3%. The argument made here is that this situation can also be represented within the model. Assuming that the household does not have access to an indexed asset while saving to buy the durable good, the household would also be paying a real interest rate of approximately 10% — that being the effect of high inflation eroding what had already been saved. With the freeze, this rate falls to around 2%. What leads households to purchase more durable goods is the reduction in the effective price, which is analogous in both cases. The only difference is that in an installment purchase, the household acquires the good as soon as it begins allocating income towards it (paying installments), while in the model the good is acquired when the household has accumulated all the necessary income (savings equal the durable good's price).

Thus, as argued here, not only cash purchases are represented within the model, but both forms of installment purchases are as well, albeit not perfectly.

5 Model

The objective of this paper is to analyze the effect of a sharp reduction in inflation on the consumption of durable goods. To do so, it is necessary to use some market imperfection so that inflation has a real effect.

First, this market imperfection is the impossibility of accessing a credit market for households, which requires consumers to save in order to purchase the durable good, thereby bearing the full effects of inflation.

Subsequently, households are given partial access to the credit market, where they can only access it after purchasing the durable good, initially paying only a portion of the good's price and financing the remainder through a perpetuity. Again, the effect of inflation is felt, though attenuated by the possibility of accessing credit after paying a down payment.

It is worth noting the rich structure of the models, as one can work with both permanent and transitory inflation reductions.

5.1 Basic Setup

The economy of both models is composed of agents who live infinitely and are heterogeneous in income, w. The possibility of heterogeneous agents in preferences will be analyzed later. N new households are born each period.

Preferences The utility function of households is given by:

$$U = \int_{t_{0i}}^{+\infty} \left[u(c_{it}) + \beta I_{it}\right] e^{-\rho(t-t_{0i})}\, dt \tag{5.1}$$

where $u' > 0$, $u'' < 0$, $u'(0) = +\infty$, ρ represents the intertemporal discount rate, β represents the utility of the durable consumer good, and t_{0i} is the period in which household i is born.

c_{it} represents the consumption of non-durable goods by household i at instant t, and I_{it} is a dummy variable equal to 1 if the household owns the durable good at instant t and 0 otherwise. Since there is no depreciation, once the durable good is purchased, it will always yield a utility of β thereafter.

5.2 No Access to the Credit Market

In this case, the market imperfection consists of the impossibility of accessing the credit market. As a result, households must save for several periods while bearing the full effects of inflation.

The constraint in equation (5.2) ensures that the real value of the individual's savings at the time of purchase equals the relative price of the durable good.

$$p_I = \int_{t_{0i}}^{T_i+t_{0i}} s_{it} e^{-\pi(T_i+t_{0i}-t)}\, dt \tag{5.2}$$

Inflation is given by π, p_I is the relative price of the durable good in relation to non-durables, T_i is the instant at which the household purchases the good, and s_{it} is the saving made at instant t by household i. Both p_I and π are exogenous.

This constraint simply states that, each period, previously saved amounts suffer erosion due to high inflation, and that the sum of all saved amounts, in real terms, must equal the value of the durable good for the household to be able to purchase it.

Finally, consumption of the non-durable good is given by equation (5.3).

$$c_{it} = \begin{cases} w_i - s_{it} & \text{if } t \in [t_{0i}, T_i + t_{0i}] \\ w_i & \text{if } t \in (T_i + t_{0i}, \infty) \end{cases} \tag{5.3}$$

where c_{it} is the consumption of the non-durable good by household i at instant t.

In addition, the household evaluates whether it is worthwhile to purchase the durable good or remain without it forever. Individual i purchases the durable good if there exist T_i and $\{s_{i,t}\}_{t_{0i}}^{\infty}$ such that:

$$\int_{t_{0i}}^{T_i+t_{0i}} u(w_i - s_{it})e^{-\rho(t-t_{0i})}dt + \int_{T_i+t_{0i}}^{+\infty} [u(w_i)+\beta]e^{-\rho(t-t_{0i})}dt > \int_{t_{0i}}^{+\infty} u(w_i)e^{-\rho(t-t_{0i})}dt \quad (5.4)$$

Equation (5.4) states that the lifetime consumption of a household that decides to purchase the durable good must yield more utility than if the good were not acquired and all income were consumed in non-durables.

Household Problem The household's problem is characterized by the following maximization:

$$\max_{\{c_{it};s_{it}\},T_i} \int_{t_{0i}}^{T_i+t_{0i}} u(c_{it})e^{-\rho(t-t_{0i})}dt + \int_{T_i+t_{0i}}^{+\infty} [u(c_{it})+\beta]e^{-\rho(t-t_{0i})}dt$$
$$\text{s.t.} \quad (5.2),\ (5.3)$$

Additionally, the household evaluates whether it is worthwhile to purchase the durable good, assessed by equation (5.4). If the optimal sequence does not satisfy equation (5.4), the household will consume all its income in non-durables, yielding utility:

$$\bar{U}_i = \int_{t_{0i}}^{+\infty} u(w_i)e^{-\rho(t-t_{0i})}dt = \frac{u(w_i)}{\rho} \quad (5.5)$$

The solution to the problem has the following first-order conditions:[7]

$$u(w_i - s_{i,T_i+t_{0i}}) - [u(w_i)+\beta] = -u'(w_i - s_{i,T_i+t_{0i}})[s_{i,T_i+t_{0i}} - \pi p_I] \quad (5.6)$$
$$\frac{u'(w_i - s_{i,T_i+t_{0i}})}{u'(w_i - s_{it})} = e^{(\rho+\pi)(T_i+t_{0i}-t)}, \quad \forall t \in [t_{0i}, T_i + t_{0i}] \quad (5.7)$$

Equation (5.6) arises from the first-order condition with respect to T_i and states that the change in household utility between the purchase period and the subsequent period must equal the change in household wealth at the time of purchase, in utility terms. That is, the loss of wealth must equal the gain in utility. Equation (5.7) is an Euler equation comparing marginal utilities between period t and period $T_i + t_{0i}$.

[7] All model derivations are presented in Appendix .

Finally, using equation (5.2), the entire saving trajectory s_{it} and the durable goods purchase time T_i can be found.

5.3 Partial Access to the Credit Market

Another possibility for adapting the model to the case of installment purchases is to allow households to pay a smaller down payment at the time of purchase and subsequently gain access to durable goods financing. This would be an approximation for the case of purchases with multiple installments. Therefore, the household problem is initially the same, with only the down payment value changing.

Let α be the share of the price paid as a down payment. From that point, the household finances the remaining value, $(1-\alpha)p_I$. This relationship is shown by:

$$\int_{T_i+t_{0i}}^{+\infty} v_{it} e^{-(r-\pi)(t-T_i-t_{0i})} dt = (1-\alpha)p_I \tag{5.8}$$

where r is the nominal interest rate of the economy, implying $(r-\pi)$ is the real interest rate. The interest rate is exogenous. v_{it} represents the installment payments by individual i at period t. Equation (5.8) shows the durable good financing contract, under which the agent commits to making payments indefinitely for the remainder of the durable good's price.

The household problem will be:

$$\begin{aligned}
\max_{\{c_{it};s_{it},v_{it},T_i\}} \quad & \int_{t_{0i}}^{T_i+t_{0i}} u(c_{it}) e^{-\rho(t-t_{0i})} dt + \int_{T_i+t_{0i}}^{+\infty} [u(c_{it})+\beta] e^{-\rho(t-t_{0i})} dt \\
\text{s.t.} \quad & \int_{t_{0i}}^{T_i+t_{0i}} s_{it} e^{-\pi(T_i+t_{0i}-t)} dt = \alpha p_I \\
& \int_{T_i+t_{0i}}^{+\infty} v_{it} e^{-(r-\pi)(t-T_i-t_{0i})} dt = (1-\alpha)p_I \\
& c_{it} = \begin{cases} w_i - s_{it} & \text{if } t \in [t_{0i}, T_i+t_{0i}] \\ w_i - v_{it} & \text{if } t \in (T_i+t_{0i}, \infty) \end{cases}
\end{aligned}$$

Assuming $\rho = r - \pi$ (i.e., the households' intertemporal rate equals the real interest rate), the first-order conditions are:

$$u(w_i - s_{i,T_i+t_{0i}}) - [u(w_i - v_{i,T_i+t_{0i}}) + \beta] = [\pi\alpha p_I - s_{i,T_i+t_{0i}}] u'(w_i - s_{i,T_i+t_{0i}}) \tag{5.9}$$

$$\frac{u'(w_i - s_{i,T_i+t_{0i}})}{u'(w_i - s_{it})} = e^{(\rho+\pi)(T_i+t_{0i}-t)}, \quad \forall t \in [t_{0i}, T_i+t_{0i}] \tag{5.10}$$

$$u'(w_i - v_{i,T_i+t_{0i}}) = u'(w_i - v_{it}) \quad \forall t \in [T_i+t_{0i}, \infty] \tag{5.11}$$

Equation (5.9) is analogous to equation (5.6) for the case without credit market access, but accounting for the benefit and cost of having the perpetuity option. Equation (5.10) is identical to equation (5.7). Finally, equation (5.11) represents an Euler equation for the perpetuity payments and yields the payment trajectory.

From equations (5.8) and (5.11), we get $v_{it} = v_{i,T_i+t_{0i}} = \rho(1-\alpha)p_I$. Using equations (5.9), (5.10), and (5.11), it is possible to find the saving trajectory, perpetuity payments, and the purchase time T_i.

Additionally, the household evaluates whether it is worthwhile to purchase the durable good, assessed by:

$$\int_{t_{0i}}^{T_i+t_{0i}} u(w_i - s_{it})e^{-\rho(t-t_{0i})}dt + \int_{T_i+t_{0i}}^{+\infty} [u(w_i - v_{i,T_i+t_{0i}}) + \beta]e^{-\rho(t-t_{0i})}dt > \int_{t_{0i}}^{+\infty} u(w_i)e^{-\rho(t-t_{0i})}dt \quad (5.12)$$

If the optimal sequence does not satisfy (5.12), the household will consume all its income in non-durables, with utility given by equation (5.5).

5.4 Permanent Stabilization

In both model variants presented so far, it is possible to introduce a permanent, credible, and unanticipated reduction in inflation from a value π_0 to π_1.[8] The government announces the permanent inflation reduction at instant t_1, and from that point consumers re-optimize their consumption and saving trajectories according to the new inflation rate.

In this case, households that have already purchased the durable good before t_1 are not affected by the regime change. Households born after the regime change optimize their problem using the new inflation rate π_1.

The main interest lies with households that have not yet purchased the durable good before the change. These can be characterized as three types: (i) households that will never purchase the good regardless of whether inflation is π_0 or π_1; (ii) the extensive margin — households that would not purchase the durable good if inflation were π_0, but start purchasing it when inflation is π_1; (iii) the intensive margin — households that were already saving for the purchase and re-optimize their trajectory with the regime change.

Extensive Margin As already noted, these households decide to purchase the durable good after the regime change. That is, with the reduction in the effective price of the durable good, it becomes worthwhile in terms of utility to purchase the good for households that previously did not. All consumption, saving, and purchase time trajectories

[8] Recall that $\pi_0 > \pi_1$.

are given by the household problem with $\pi = \pi_1$. These households represent the largest share of the increase in durable goods consumption in the model, as will be presented subsequently.

Intensive Margin In this case, households were already saving before the regime change and will re-optimize their saving and consumption trajectories. The optimal choice will still be to purchase the durable good.

The new problem for these households, in the case without credit access, will be:

$$\begin{aligned}
\max_{\{c_{it};s_{it},T_i'\}} \quad & \int_{t_1}^{T_i'+t_1} u(c_{it})e^{-\rho(t-t_1)}dt + \int_{T_i'+t_1}^{+\infty} [u(c_{it})+\beta]e^{-\rho(t-t_1)}dt \\
\text{s.t.} \quad & p_I = \int_{t_{0i}}^{t_1} \bar{s}_{it}e^{-\pi_0(t_1-t)-\pi_1 T_i'}dt + \int_{t_1}^{T_i'+t_1} s_{it}e^{-\pi_1(T_i'+t_1-t)}dt \\
& c_{it} = \begin{cases} w_i - s_{it} & \text{if } t \in [t_1, T_i'+t_1] \\ w_i & \text{if } t \in (T_i'+t_1, \infty) \end{cases}
\end{aligned}$$

where $\{\bar{s}_{it}\}_{t=t_{0i}}^{t_1}$ are the amounts already saved up to that point, and T_i' is the new purchase time.

In both models, the effect of reduced inflation has two sides. On one hand, the reduction in inflation generates a decrease in the effective price of the consumer good, which would stimulate consumption and tend to reduce the durable goods purchase time. On the other hand, inflation reduces the cost of maintaining savings, which causes households to tend to delay their durable goods consumption more. These two effects counteract each other, and only with parameter values can one assess which effect dominates.

5.5 Temporary Stabilization

In this case, there is an unanticipated reduction in inflation that will last only for a certain period. That is, a reduction in inflation from π_0 to π_1 occurs at period t_1 and lasts until period t_2, when inflation increases to π_2.[9]

Again, for households that purchased the good before t_1, there is no change. For households born after t_2, or for households born between t_1 and t_2 that purchase the durable good before t_2, the problem is the same, using inflation π_2 and π_1, respectively.

Extensive Margin Movements on this margin can be of two types. The first consists of households that manage to purchase the good before t_2. For these households, the case is identical to the permanent stabilization case.

[9]This may or may not equal π_0.

The second type concerns households that can only acquire the durable good after period t_2. Again, households will purchase the good if and only if:

$$\int_{t_1}^{T_i+t_1} u(w_i - s_{it})e^{-\rho(t-t_1)}dt + \int_{T_i+t_1}^{+\infty} [u(w_i) + \beta]e^{-\rho(t-t_1)}dt > \int_{t_1}^{+\infty} u(w_i)e^{-\rho(t-t_1)}dt \quad (5.13)$$

In addition, the saving and consumption strategy satisfies:

$$c_{it} = \begin{cases} w_i - s_{it} & \text{if } t \in [t_1, T_i + t_1] \\ w_i & \text{if } t \in (T_i + t_1, \infty) \end{cases} \quad (5.14)$$

$$p_I = \int_{t_1}^{t_2} s_{it}e^{-\pi_1(t_2-t_1)-\pi_2(T_i+t_1-t_2)}dt + \int_{t_2}^{T_i+t_1} s_{it}e^{-\pi_2(T_i+t_1-t)}dt \quad (5.15)$$

Intensive Margin As with the extensive margin, the intensive margin has two cases. First, households that purchase before t_2 face the same problem as in the permanent transition case and therefore will not be re-presented. The second case considers households that can only purchase the durable good after t_2. The problem for these households is:

$$\begin{aligned} \max_{\{c_{it};s_{it},T_i'\}} \quad & \int_{t_1}^{T_i'+t_1} u(c_{it})e^{-\rho(t-t_1)}dt + \int_{T_i'+t_1}^{+\infty} [u(c_{it}) + \beta]e^{-\rho(t-t_1)}dt \\ \text{s.t.} \quad & p_I = \int_{t_{0i}}^{t_1} \bar{s}_{it}e^{-\pi_0(t_1-t)-\pi_1(t_2-t_1)-\pi_2(T_i'+t_1-t_2)}dt \\ & \quad + \int_{t_1}^{t_2} s_{it}e^{-\pi_1(t_2-t)-\pi_2(T_i'+t_1-t_2)}dt + \int_{t_2}^{T_i'+t_1} s_{it}e^{-\pi_2(T_i'+t_1-t)}dt \\ & c_{it} = \begin{cases} w_i - s_{it} & \text{if } t \in [t_1, T_i' + t_1] \\ w_i & \text{if } t \in (T_i' + t_1, \infty) \end{cases} \end{aligned}$$

6 Simulation

This section presents the simulation of both models, analyzing the effect of the inflation reduction from the Cruzado Plan. Simulations are performed with permanent stabilizations in both model frameworks and temporary simulations for the case in which households do not have access to the credit market.

All simulations share the following characteristics:

- N households are born each period.

- The economy operates with inflation π_0 for t_1 periods.
- From t_1 onwards, inflation is reduced to π_1.

The first subsection presents the calibration of the model parameters. Subsequent subsections present and compare the main simulation results.

6.1 Parameter Calibration

The model was calibrated so that each time period represents one month. The base period for calibration was September 1985, which is the date of the PNAD closest to the Cruzado Plan, providing the dataset most proximate to the price freeze.

The price of durable goods was extracted from the refrigerator price series constructed from newspaper advertisements and deflated using the INPC to make prices comparable. The proxy used for the non-durable goods basket price was the cost of the basic consumption basket calculated by DIEESE in September 1985. Only the São Paulo price was used. First, it is the city with the largest weight in national consumption. Second, with the exception of Salvador, price differences between capital cities are not large. Third, using all seven cities would raise the question of how to weight each to calculate the average basket price. Thus, since durable goods prices were taken from advertisements published in São Paulo, only the São Paulo basic basket price was used.

To construct p_I, the median of the price series was used, divided by the basic basket price. Real household income was generated randomly from a log-normal distribution, with mean and variance parameters taken from the household income series from the 1985 PNAD divided by the basic basket price in the same period.

The intertemporal discount rate ρ was set at 0.01, implying a real interest rate of 1% per month. This value was set based on the real interest rate for 1985.[10] The parameter β — the marginal utility of the durable good — was set at 0.0374 so that the simulated share of households with durable goods before the shock would match the 1985 PNAD figure.[11]

The inflation rates π_0, π_1, and π_2 used are approximately the monthly inflation rates before and during the Cruzado Plan, with π_2 initially equal to π_0.[12] Although π_2 is lower than the recorded level, the simulation was set so that post-plan inflation returned to the same level, with other values being tested in the robustness analysis.

The value of t_1 is set high enough to allow the economy to reach steady state, while the values of t_2 were varied to simulate different possible plan durations.

Table 4 summarizes the main values used in the simulations.

[10] The average for the period was 0.96%, considering the real interest rate as the monthly Selic rate minus the INPC.

[11] Only the refrigerator ownership share in 1985 was considered. This share was 63.9%.

[12] The monthly inflation averages measured by INPC were 11.16% for 1985, 2.12% for the period from March to December 1986, and 14.38% for 1987.

Table 4: Main Simulation Parameters

p_I	w_μ	w_{σ^2}	β	ρ	π_0	π_1	π_2	N
5.22	7.05	126.13	0.0374	0.01	0.1	0.02	0.1	100

Source: Own elaboration.

Finally, it is worth noting that the utility function used in all simulations is the logarithmic function, i.e., $u(c) = \log(c)$.

6.2 Permanent Shock

No credit access First, the simulation was run for the model in which households have no access to the credit market and can only purchase the durable good by saving over a period during which the saved amount is not protected from inflation. Figure 2 shows the impact of price stabilization at $t = 0$ on durable goods ownership.

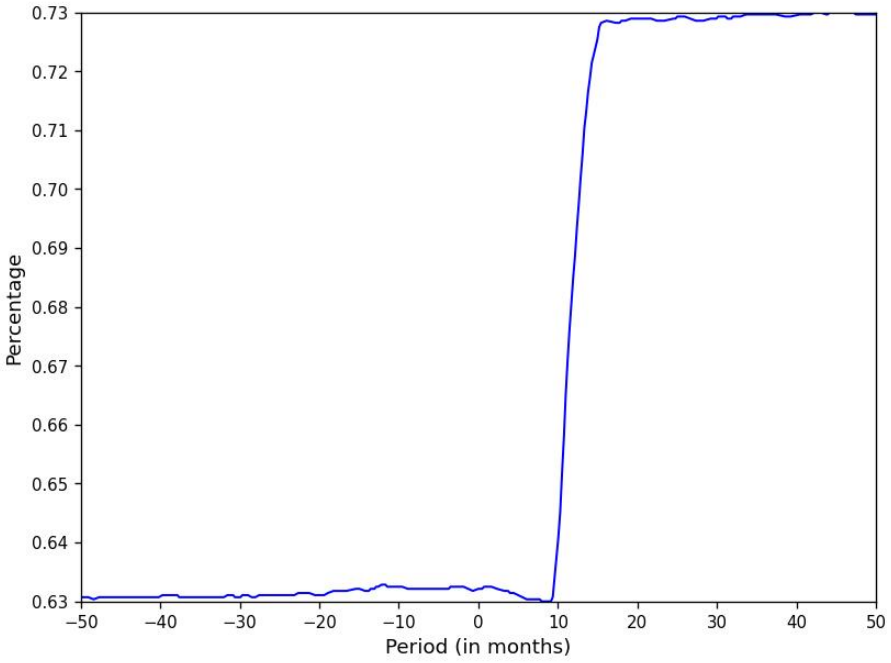


Figure 2: Percentage of households with durable good before and after the shock.
Source: Own elaboration.

The effect is approximately 5 pp in the percentage of households with the durable good after 1 year. After 2 years, the effect is approximately 8 pp. It is also possible to note in the figure that the increase occurs approximately 10 periods after the freeze.

This is because the largest part of the effect comes from the fourth income quintile, i.e., the extensive margin. The average saving time for these households is 11.56 periods, thus generating this 10-period lag.

Figure 3 shows the decomposed effect between the extensive and intensive margins.

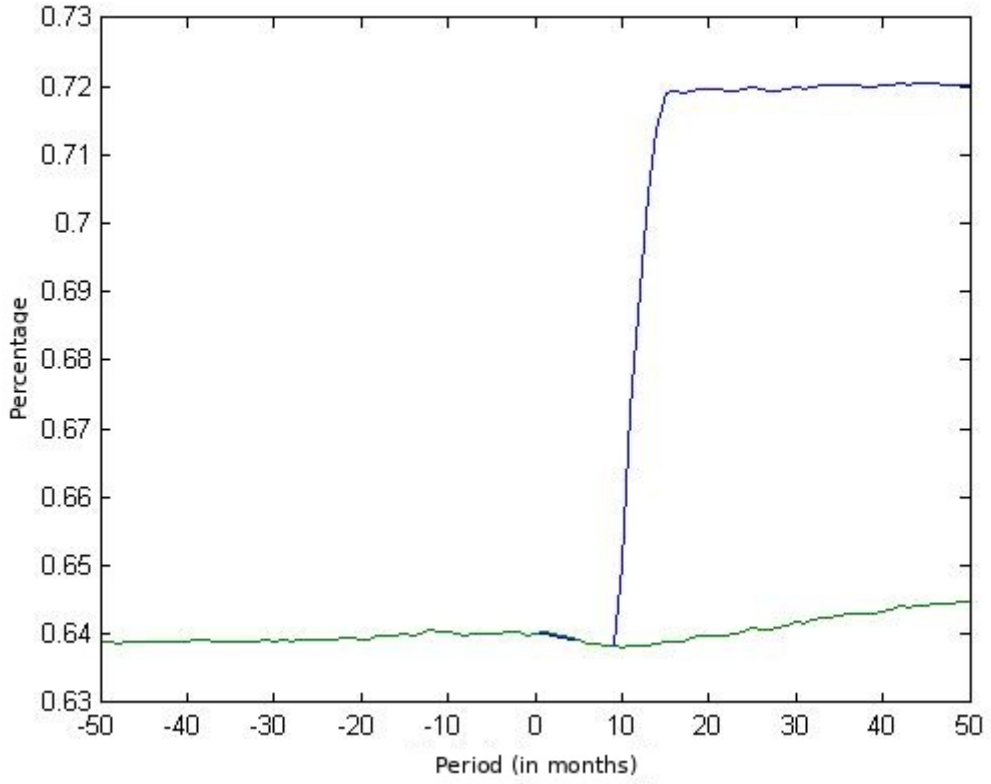


Figure 3: Decomposed impact of the shock (extensive and intensive margins)
Source: Own elaboration.

The blue line maintains intensive margin households as if they did not have the effect of the inflation reduction, thus representing only the extensive margin effect. The green line shows only the intensive margin effect, holding inflation constant for extensive margin households. As can be seen, the effect comes almost entirely from the extensive margin.

Table 5: Average saving time (in months)

	Before shock	After shock
Mean	3.30	8.32
Maximum	8.09	15.26
Minimum	0.08	0.01

Source: Own calculations.

Another important point to highlight is the greater preference for consumption smoothing among households. Table 5 illustrates the average saving time (in months) of households that purchase the durable good, along with the maximum and minimum values. The increase in saving time after the shock is pronounced, as the mean rose from approximately 3 months and 9 days to approximately 8 months and 9 days.

Table 6: Average household saving time (in months)

Margin	Average time
Extensive	11.56
Intensive	3.89
Remaining households before shock	3.30
Remaining households after shock	4.92

Source: Own calculations.

Table 6 shows the average saving time (in months) for the intensive and extensive margins compared with the rest of the population before and after the shock. From the table, all households smooth consumption relative to prior behavior. Extensive and intensive margin households have average saving times of 11 months and 17 days and 3 months and 27 days, respectively, while households before the inflation reduction had an average saving time of 3 months and 9 days.

This result of increased saving time indicates that the consumption smoothing effect dominates in the case of a pronounced and permanent reduction in inflation. This implies that the smoothing effect is larger than the effect of the fall in the effective price of the durable good. That is, the fall in inflation increases the incentive for individuals to purchase faster due to the reduction in the effective price of the good, but the preference for consumption smoothing and the lower erosion of savings leads households to increase their saving time.

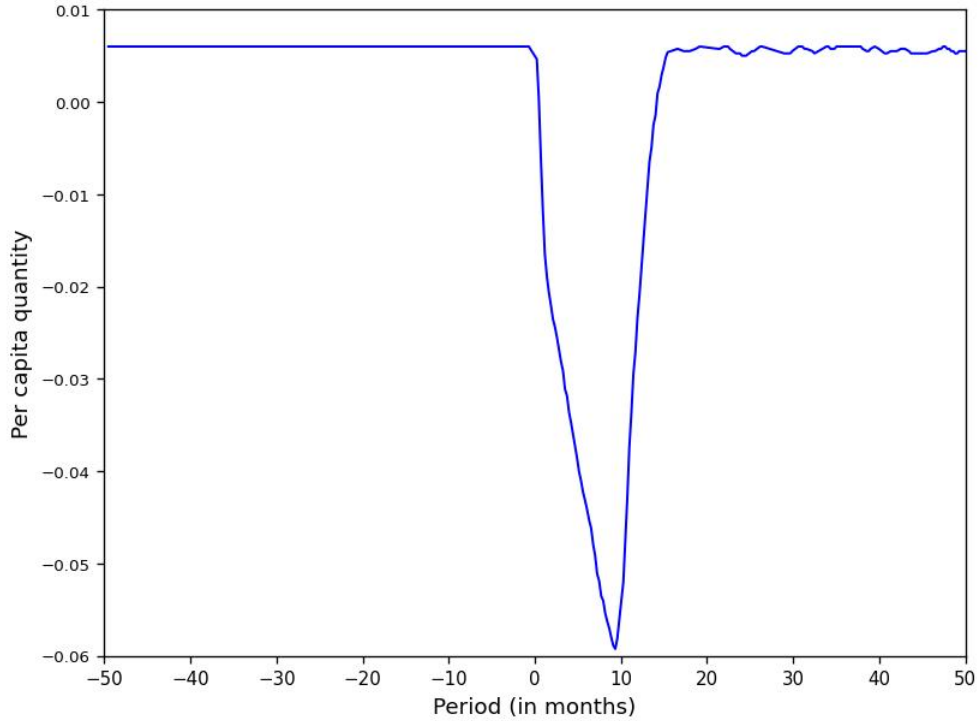


Figure 4: Difference in per capita consumption of non-durable goods
Source: Own calculations.

Finally, Figure 4 shows the difference in per capita consumption of non-durable goods between the price stabilization scenario and the high inflation scenario. Two behaviors are clear. First, there is a sharp fall in consumption during the first 10 months — the period during which the extensive margin is saving. After the 10th month, consumption returns to its original level relative to the high inflation scenario and becomes slightly higher from the 13th period after the shock onward.

These two facts arise from the interaction of two distinct effects. On one hand, the increase in the number of saving households causes a fall in non-durable goods consumption in the early periods, driven by the extensive margin. On the other hand, the greater consumption smoothing explains the slight increase in non-durable goods consumption after the initial shock stabilizes, i.e., after the 10th period.

Partial credit market access The simulation was also run for the case in which households have partial credit market access — that is, they can purchase the durable good by saving over a period and paying a down payment, with saved amounts not protected from inflation. After this down payment of αp_I, the individual finances the rest of the durable good's price through a perpetuity. Initially, $\alpha = 0.33$.

Figure 5 shows the impact of price stabilization at instant 0 on durable goods ownership.

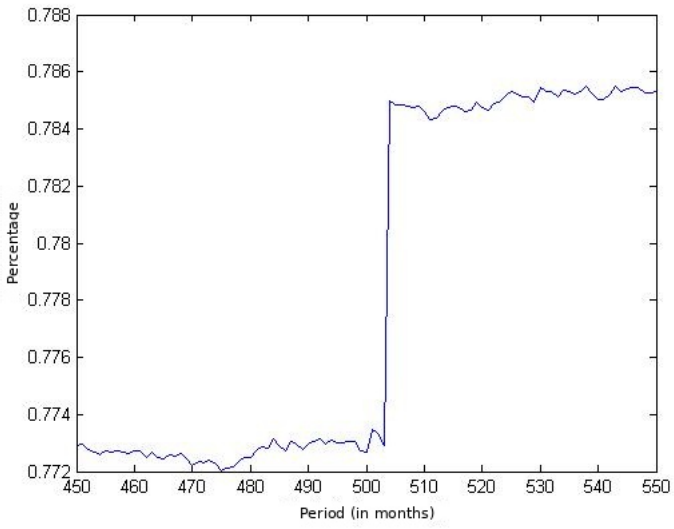


Figure 5: Percentage of households with durable good before and after the shock
Source: Own elaboration.

The effect is approximately 1 pp in the percentage of households with the durable good. The increase occurs approximately 3 periods after the freeze. The effect originates from the extensive margin, which has an average saving time of 3.14 months, generating the 3-period lag. Again, the largest part of the extensive margin comes from the fourth quintile.

It is worth noting that the initial share of households owning the durable good before the shock is higher than in the previous simulation (approximately 77%), since it is easier to purchase the good due to partial credit market access and the possibility of financing $(1 - \alpha)$ of the good's value.

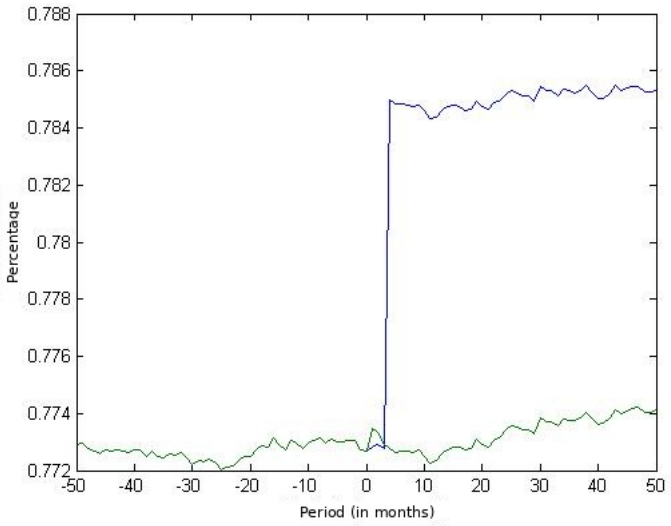


Figure 6: Decomposed impact of the shock (extensive and intensive margins
Source: Own elaboration.

Again, the effect comes almost entirely from the extensive margin.

Table 7: Average saving time (in months) — partial credit

	Before shock	After shock
Mean	1.68	2.27
Maximum	5.25	9.19
Minimum	0.02	0.01

Source: Own calculations.

Table 7 shows that the increase in average saving time after the shock is pronounced, with the mean rising from 1 month and 20 days to approximately 2 months and 8 days.

Table 8: Average household saving time (in months) — partial credit

Margin	Average time
Extensive	3.14
Intensive	2.36
Remaining households before shock	1.68
Remaining households after shock	2.18

Source: Own calculations.

Table 8 shows the average saving time for the intensive and extensive margins compared with the rest of the population. All households smooth consumption relative to prior behavior. Extensive and intensive margin households have average saving times of 3 months and 4 days and 2 months and 11 days, respectively. Households appearing before and after the inflation reduction had average saving times of 1 month and 20 days and 2 months and 5 days, respectively.

This result again indicates that the consumption smoothing effect dominates in the case of a pronounced and permanent inflation reduction.

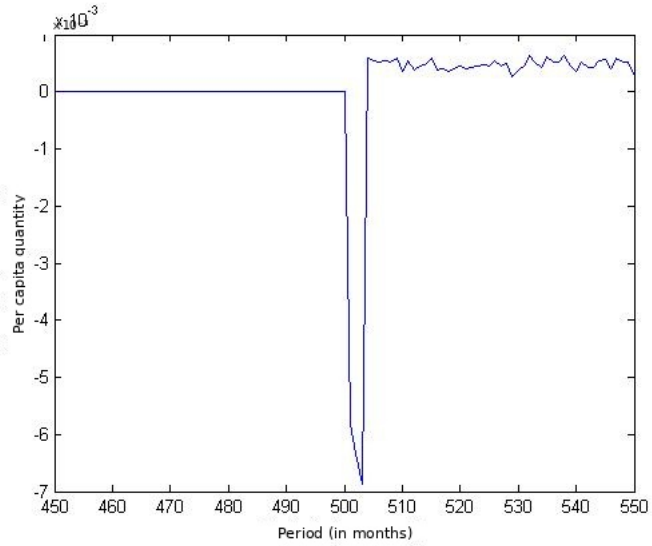


Figure 7: Difference in per capita consumption of non-durable goods — partial credit
Source: Own elaboration.

Figure 7 shows two clear behaviors: first, a sharp fall in consumption during the first 3 months (when the extensive margin is saving); then, after the 4th month, consumption begins to recover relative to the high inflation scenario. These two facts arise from the interaction of two effects. On one hand, the increase in the number of saving households causes a fall in non-durable consumption in the early periods, driven by the extensive margin. On the other hand, the recovery occurs as the extensive margin movement ends.

6.3 Temporary Shock

This section presents the simulation for the case in which households have no credit market access and can only purchase the durable good by saving over a period during which savings are not protected from inflation. However, the inflation reduction is temporary. The period in which inflation returns to its previous level was calibrated to match the Cruzado Plan's duration of 10 months. That is, t_2 — the time at which inflation returns to its prior level ($\pi_2 = \pi_0$) — was set at 510. Figure 8 shows the impact of price stabilization from instant 0 to instant 10 on durable goods ownership.

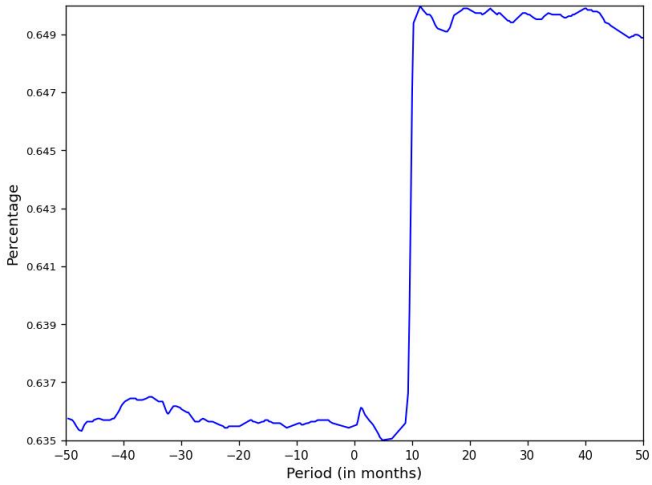


Figure 8: Percentage of households with durable good before and after the shock
Source: Own elaboration.

The effect was approximately 1.26 pp in the share of households with the durable good after 1 year. It is also possible to note that the increase occurs approximately 10 periods after the freeze. The average saving time of the extensive margin is 9.78 periods, generating this 10-period lag. The largest part of the effect comes from the extensive margin.

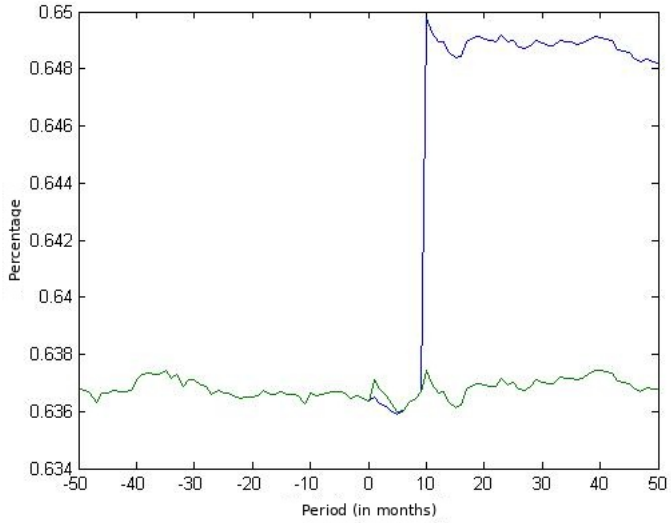


Figure 9: Decomposed impact of the shock (extensive and intensive margins)
Source: Own elaboration.

Table 9: Average saving time (in months) — temporary shock

	Before shock	During shock	After shock
Mean	3.29	6.33	3.44

Source: Own calculations.

Table 9 illustrates the average saving time (in months). The increase after the shock is pronounced: higher during the shock period and decreasing again after the inflation reduction ends. The mean went from approximately 3 months and 9 days to 6 months and 10 days during the freeze, and to 3 months and 13 days after the freeze.

Table 10: Average saving time by margin (in months) — temporary shock

Margin	Average time (months)
Extensive	9.78
Intensive	5.66
Remaining households before shock	3.29
Remaining households after shock	3.35

Source: Own calculations.

Table 10 confirms the increase in saving time for both the extensive and intensive margins, as well as for the remaining households purchasing after the shock.

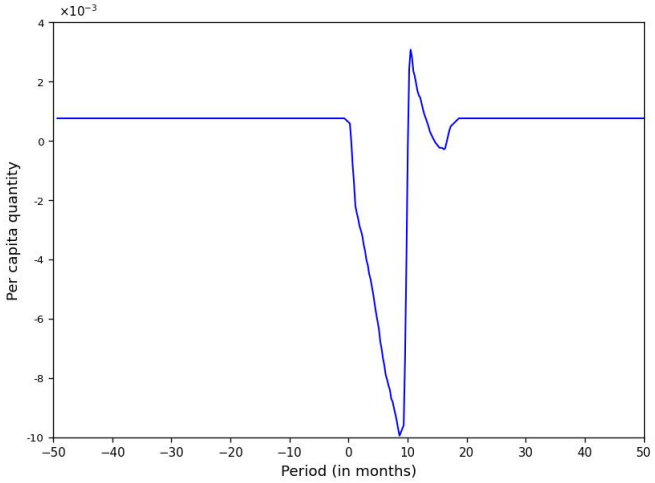


Figure 10: Difference in per capita consumption of non-durable goods — temporary shock
Source: Own elaboration.

Figure 10 indicates that non-durable goods consumption initially falls, subsequently recovering and maintaining the same level as before the shock.

6.4 Comparison with Data

Table 11: Durable goods consumption expansion in one year (in pp)

	No credit access		Partial credit	PNAD
	Permanent	Temporary	Permanent	
PNAD	5.19	1.26	1.17	1.8

Source: Own calculations. Data: PNAD.

From Table 11, the permanent shock simulation without credit market access overestimates the shock, while the remaining simulations underestimate it.

Beyond the magnitude of the expansion, it is worth noting that in all simulations the most affected quintile is the fourth. In the PNAD, the largest expansion comes precisely from this income quintile, reaching 5 pp in the data without any controls. In the closest simulation — the temporary shock — the effect in the fourth quintile is an expansion of 6.27 pp, which closely matches the PNAD data. That is, the model can explain the asymmetry between quintiles: the expansion is larger in the fourth quintile precisely because it has more households in the extensive margin that only marginally prefer not to acquire the durable good during high inflation.

The simulation that most closely matches the observed impact is the one with a temporary shock and households without credit access. For this reason, the possibility of preference heterogeneity will be introduced in this simulation in the robustness analysis.

7 Robustness Analysis

Before concluding, this section discusses the sensitivity of the results to the parameterization chosen for the simulations. New simulations of permanent reduction are run to assess the impact of parameters β, p_I, and ρ.

In addition, the impact of varying α will be assessed for the case of permanent stabilization with partial credit market access.

Finally, the possibility of assigning a preference parameter to each household — making households heterogeneous in their preferences — will be presented.[13]

Tables 12 and 13 show the main results of the robustness analysis. Bold values indicate the parameters that were changed in each simulation.

Table 12: Robustness analysis and impact on durable goods ownership

β	ρ	p_I	Shock 12m (%)	Shock 50m (%)	Initial share (%)
0.0374	0.01	5.22	5.19%	8.22%	63.87%
0.01	0.01	5.22	4.51%	4.56%	28.33%
0.10	0.01	5.22	0.04%	7.85%	82.08%
0.0374	0.005	5.22	0.00%	8.26%	78.12%
0.0374	0.05	5.22	2.62%	2.64%	20.24%
0.0374	0.01	2.50	3.13%	4.93%	84.26%
0.0374	0.01	10.00	5.88%	9.36%	41.32%

Source: Own calculations.

Table 13: Robustness analysis and average saving time

β	ρ	p_I	Before shock	After shock
0.0374	0.01	5.22	3.29	8.27
0.01	0.01	5.22	2.34	5.37
0.10	0.01	5.22	3.79	10.22
0.0374	0.005	5.22	4.13	11.40
0.0374	0.05	5.22	1.51	2.83
0.0374	0.01	2.50	2.48	6.22
0.0374	0.01	10.00	3.95	9.52

Source: Own calculations.

[13]It is possible to also make them heterogeneous with respect to price and the intertemporal discount rate. However, these two hypotheses seem less realistic, and for this reason no simulations were run in this direction.

For the marginal utility of the durable good, the baseline is 0.0374, and two additional values were tested: 0.01 and 0.1. As shown in Table 12, an increase in β raises the desirability of the durable good. Holding other parameters constant, the share of households acquiring the durable good even with high inflation increases with β.

Regarding the shock, an inverted U-shape pattern appears: the shock magnitude is initially smaller, grows to the calibrated value, and then decreases again beyond a certain value. This can be seen in the shock effect at 50 months. The explanation is that for low values of β few households initially own the durable good. As β increases, it becomes more attractive for a marginal household to purchase the durable good, given its increased marginal utility. Beyond a certain point, the share of households is large enough that fewer households are on the extensive margin, thereby reducing the impact of the inflation reduction.

In Table 13, average saving time clearly increases with the marginal utility of the durable good. This effect is generated by the poorer households that start saving as well, and given their lower income, they necessarily save for longer, raising the aggregate statistic.

For the impatience rate, the following values were tested: 0.005, 0.01, and 0.05. The baseline value was $\rho = 0.01$. A higher impatience rate means the present is more highly valued, and forgoing part of the endowment to obtain the durable good becomes more costly for households. In addition, the saving trajectory becomes less smooth, with an increasing share of savings made close to the purchase period. The result, as shown in Table 12, is a reduction in the share of households acquiring the good with high inflation as ρ increases.

The reduction in average saving time — shown in Table 13 — occurs both because only wealthier households purchase the durable good for higher values of ρ (which itself leads to a reduction) and because of the greater valuation of the present, which leads those who purchase the durable good to do so even earlier.

The implications of increasing the relative price of the durable good are the most straightforward to interpret. A higher relative price means the household must forgo increasingly more non-durable goods, causing an ever-smaller share of households to acquire the good — as illustrated in Table 12. The simulated relative prices were 2.5, 5.22, and 10.

With a higher price, the amount that must be saved to acquire the good also increases, which also increases (ceteris paribus) the average saving time, as shown in Table 13. What is most striking is the increase in the shock's impact as p_I increases. Since fewer households initially own the good, the extensive margin will be larger.

7.1 Down Payment Share (α)

For the case of partial credit market access, the consequences of varying the down payment value (α) are small. Table 14 presents the initial share of households with the durable good before the shock and the change caused by the shock.

Table 14: Households with durable good immediately before t_1 (%) and 12-month shock impact (%)

	$\alpha = 0.125$	$\alpha = 0.33$	$\alpha = 0.5$
Initial share	79.61%	77.27%	74.13%
Shock impact	0.00%	1.17%	2.95%

Source: Own calculations.

Note that the higher the down payment value, the lower the initial share of households with the durable good, since fewer households can save long enough to pay the down payment. On the other hand, the shock effect intensifies as the down payment increases. This occurs because the smaller the down payment, the more households can finance the good and access the credit market, which reduces the extensive margin effect.

Table 15: Average saving time (in months) by α

	$\alpha = 0.125$	$\alpha = 0.33$	$\alpha = 0.5$
Before shock	0.79	1.68	2.21
After shock	0.97	2.27	3.43

Source: Own calculations.

The smaller the down payment, the less time individuals save, since they can accumulate the down payment value faster and thus finance the remainder through the perpetuity.

7.2 Preference Heterogeneity

This subsection discusses the possibility of introducing preference heterogeneity among households. The main objective is to better distribute the shock across all five income quintiles so that it is not concentrated only among the fourth and third quintile households, as in previous models.

For this purpose, the temporary 10-period simulation without credit access was used as the baseline.

First, preferences were simulated with β distributed from a truncated normal at zero, with various mean and variance parameters. The main results are presented in Table 16.

Table 16: Main results of preference heterogeneity

	Shock impact (%)					
	$\sigma_\beta = 0.005$	$\sigma_\beta = 0.006$	$\sigma_\beta = 0.007$	$\sigma_\beta = 0.008$	$\sigma_\beta = 0.009$	$\sigma_\beta = 0.010$
$\mu_\beta = 0.035$	1.76%	1.97%	1.98%	2.09%	2.09%	2.20%
$\mu_\beta = 0.036$	1.68%	1.70%	1.67%	1.94%	1.85%	1.96%
$\mu_\beta = 0.037$	1.46%	1.56%	1.63%	1.68%	1.75%	1.93%
$\mu_\beta = 0.038$	1.27%	1.34%	1.37%	1.49%	1.55%	1.83%
$\mu_\beta = 0.039$	1.23%	1.10%	1.28%	1.26%	1.46%	1.55%
$\mu_\beta = 0.040$	0.92%	1.08%	1.04%	1.31%	1.28%	1.29%
	Households with durable good at t_1 (%)					
	$\sigma_\beta = 0.005$	$\sigma_\beta = 0.006$	$\sigma_\beta = 0.007$	$\sigma_\beta = 0.008$	$\sigma_\beta = 0.009$	$\sigma_\beta = 0.010$
$\mu_\beta = 0.035$	61.79%	61.88%	61.66%	61.44%	60.99%	60.83%
$\mu_\beta = 0.036$	62.85%	62.17%	62.42%	62.26%	61.81%	61.40%
$\mu_\beta = 0.037$	63.68%	62.92%	63.23%	62.74%	62.53%	62.36%
$\mu_\beta = 0.038$	63.95%	63.96%	63.83%	63.54%	63.37%	63.11%
$\mu_\beta = 0.039$	64.42%	64.83%	64.38%	64.26%	63.78%	63.54%
$\mu_\beta = 0.040$	65.56%	64.79%	65.10%	64.87%	64.38%	64.35%

Source: Own calculations.

Note that fixing μ_β and increasing σ_β raises the shock's impact on the share of households with durable goods. In addition, fixing σ_β and increasing μ_β raises the share of households with the durable good in the period immediately before the shock begins.

Table 16 also shows that when $\mu_\beta = 0.038$ and $\sigma_\beta = 0.01$, the shock's impact on durable goods consumption is 1.83% and the share of households with the durable good immediately before the shock begins is 63.11%. This is the simulation that most closely matches the PNAD data.

Table 17 shows the income quintiles for each one of the cases.

Table 17: Share of durable goods ownership by income quintile

	5th Quintile	4th Quintile	3rd Quintile	2nd Quintile	1st Quintile
PNAD					
1985	22.6%	44.6%	66.6%	84.9%	96.7%
1986	25.5%	49.5%	68.0%	86.5%	96.8%
Temporary simulation					
Before	0.0%	19.8%	99.1%	99.6%	99.8%
After	0.0%	26.0%	99.2%	99.5%	99.8%
Heterogeneous β					
Before	10.2%	44.5%	55.9%	60.5%	63.2%
After	10.2%	47.6%	57.8%	60.8%	63.5%

Source: Own calculations.

Comparing the data, one can verify that the simulation with heterogeneous β is closer to the data, making it advantageous for better explaining inflation reduction episodes.

8 Conclusion

This paper examined the consumption expansion that follows price stabilizations via price or exchange rate controls. Unlike most of the literature, the approach employed focused on durable goods consumption — generally neglected in the literature — combined with a restriction on access to indexed assets.

Compared to Helpman and Razin (1987) and Rebelo (1997), the model has the advantage that the shock does not depend on additional government actions beyond controlling inflation. Unlike Calvo (1986), even fully credible plans still generate an expansion. Furthermore, in contrast to De Gregorio et al. (1998), the model clearly specifies the channel linking price stabilization to the expansion in durable goods consumption, and the effect of stabilization on the intensive margin is not to frontload purchases but rather to cause households to smooth consumption more, increasing average saving time.

Simulation results point to an exaggerated effect of price reduction on durable goods ownership in the case of permanent stabilization, both in the absence of credit access and in the case of partial access. A proposed explanation is that the model assumes the success probability of the shock is 100% in this case, which exaggerates the expected reduction in the good's effective price.

For temporary stabilizations, it was shown that the magnitude of the expansion is somewhat smaller than observed in the data. A weakness of the temporary shock analysis is that, while the permanent shock implicitly assumes that the success probability of the

stabilization plan is 100%, the temporary shock assumes this probability is zero from a certain point (t_2) onwards.

It is worth noting that the model can be calibrated in various ways and has a wide range of possible variations. Among these is the possibility of differentiating households by both income and preferences. The result was that the model better predicts the data when this preference heterogeneity is exploited.

Another point deserving greater attention in future work is the dynamics of non-durable goods consumption. At present, the effect of stabilization in this market is a reduction in non-durable consumption initially — due to all households that will start saving given the reduction in the effective price of the durable good — followed by an increase in per capita consumption driven by greater consumption smoothing. Perhaps the addition of a labor market could reduce this initial contraction, as the increased demand for durable goods would also positively affect household income.

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Appendix

This appendix presents some results of the third chapter of this thesis. In particular, the models used for the simulations are shown, already using the utility function $u(c) = \log(c)$. In addition, the tables related to the linear probability model and a section with the logic of the program used for the simulations are presented.

Household Problem

This case represents households that are not affected by the shock and the extensive margin households in the permanent shock — in which π_0 is replaced by π_1 — or the extensive margin in the temporary shock where the household purchases before t_2 and π_0 is also replaced by π_1.

Solution of the model without credit access

$$L = \int_{t_{0i}}^{T_i+t_{0i}} \log(w_i - s_{it}) e^{-\rho(t-t_{0i})} dt + \int_{T_i+t_{0i}}^{+\infty} [\log(w_i) + \beta] e^{-\rho(t-t_{0i})} dt + \lambda \left[\int_{t_{0i}}^{T_i+t_{0i}} s_{it} e^{-\pi_0(T_i+t_{0i}-t)} dt - p_I \right]$$

The first-order conditions are:

$$\begin{aligned} T_i: &\quad e^{-\rho T_i}\{\log(w_i - s_{T_i+t_{0i}}) - [\log(w_i) + \beta]\} + \lambda[s_{T_i+t_{0i}} - \pi_0 p_I] = 0 \\ s_{it}: &\quad -\frac{e^{-\rho(t-t_{0i})}}{w_i - s_{it}} + \lambda e^{-\pi_0(T_i+t_{0i}-t)} = 0, \quad \forall t \in [t_{0i}, T_i + t_{0i}] \end{aligned}$$

The first-order condition for s_{it} evaluated at $t = T_i + t_{0i}$ implies:

$$\lambda = \frac{e^{-\rho T_i}}{w_i - s_{T_i+t_{0i}}}$$

With this multiplier, the first-order conditions become:

$$\log(w_i - s_{i,T_i+t_{0i}}) - [\log(w_i) + \beta] = \frac{\pi_0 p_I - s_{i,T_i+t_{0i}}}{w_i - s_{i,T_i+t_{0i}}}$$

$$s_{it} = w_i - (w_i - s_{i,T_i+t_{0i}}) e^{(\rho+\pi)(T_i+t_{0i}-t)}$$

Using equation (3.2) — the constraint that the good's price must equal the total accumulated savings discounted for inflation — we arrive at:

$$\log(w_i - s_{i,T_i+t_{0i}}) - [\log(w_i) + \beta] = \frac{\pi_0 p_I - s_{i,T_i+t_{0i}}}{w_i - s_{i,T_i+t_{0i}}}$$

$$s_{i,T_i+t_{0i}} = w_i - \frac{\rho}{e^{\rho T_i} - 1}\left[\frac{w_i}{\pi_0}(1 - e^{-\pi_0 T_i}) - p_I\right]$$

From these two equations, T_i and $s_{i,T_i+t_{0i}}$ can be found.

Solution of the model with partial credit access

The Lagrangian for this problem is:

$$L = \int_{t_{0i}}^{T_i+t_{0i}} \log(w_i - s_{it}) e^{-\rho(t-t_{0i})} dt + \int_{T_i+t_{0i}}^{+\infty} [\log(w - v_{it}) + \beta] e^{-\rho(t-t_{0i})} dt$$
$$+ \lambda\left[\int_{t_{0i}}^{T_i+t_{0i}} s_{it} e^{-\pi(T_i+t_{0i})} dt - \alpha p_I\right] + \mu\left[\int_{T_i+t_{0i}}^{+\infty} v_{it} e^{-(r-\pi)(t-T_i-t_{0i})} dt - (1-\alpha) p_I\right]$$

The first-order conditions are:

$$T_i: \quad e^{-\rho T_i}\{\log(w_i - s_{T_i+t_{0i}}) - [\log(w_i - v_{T_i+t_{0i}}) + \beta]\} = \lambda[\pi\alpha p_I - s_{T_i+t_{0i}}] + \mu[v_{T_i+t_{0i}} - \rho(1-\alpha)p_I]$$

$$s_{it}: \quad -\frac{e^{-\rho(t-t_{0i})}}{w_i - s_{it}} + \lambda e^{-\pi(T_i+t_{0i}-t)} = 0, \quad \forall t \in [t_{0i}, T_i + t_{0i}]$$

$$v_{it}: \quad -\frac{e^{-\rho(t-t_{0i})}}{w_i - v_{it}} + \mu e^{-(r-\pi)(t-T_i-t_{0i})} = 0, \quad \forall t \in [T_i + t_{0i}, +\infty)$$

The first-order conditions for s_{it} and v_{it} evaluated at $t = T_i + t_{0i}$ imply:

$$\lambda = \frac{e^{-\rho T_i}}{w_i - s_{T_i+t_{0i}}}, \qquad \mu = \frac{e^{-\rho T_i}}{w_i - v_{T_i+t_{0i}}}$$

With both multipliers, the first-order conditions become:

$$\log(w_i - s_{i,T_i+t_{0i}}) - [\log(w_i - v_{i,T_i+t_{0i}}) + \beta] = \frac{v_{T_i+t_{0i}} - \rho(1-\alpha)p_I}{w_i - v_{T_i+t_{0i}}} + \frac{\pi\alpha p_I - s_{i,T_i+t_{0i}}}{w_i - s_{i,T_i+t_{0i}}}$$

$$s_{it} = w_i - (w_i - s_{i,T_i+t_{0i}}) e^{(\rho+\pi)(T_i+t_{0i}-t)}$$

$$v_{it} = w_i - (w_i - v_{i,T_i+t_{0i}}) e^{(\rho-(r-\pi))(T_i+t_{0i}-t)}$$

Assuming $\rho = r - \pi$, we get $v_{it} = v_{i,T_i+t_{0i}}$. Using the perpetuity constraint (3.8) and constraint (3.2):

$$\log(w_i - s_{i,T_i+t_{0i}}) - [\log(w_i - v_{i,T_i+t_{0i}}) + \beta] = \frac{\pi\alpha p_I - s_{i,T_i+t_{0i}}}{w_i - s_{i,T_i+t_{0i}}}$$
$$s_{i,T_i+t_{0i}} = w_i - \frac{\rho}{e^{\rho T_i} - 1}\left[\frac{w_i}{\pi}(1 - e^{-\pi T_i}) - \alpha p_I\right]$$
$$v_{i,T_i+t_{0i}} = (1-\alpha)p_I\rho$$

From these three equations, T_i, $s_{i,T_i+t_{0i}}$, and $v_{i,T_i+t_{0i}}$ can be found.

Solution of the model with permanent shock for intensive margin households

This case represents the intensive margin households in the permanent shock, as well as in the temporary shock for $T_i \leq t_2$.

$$L = \int_{t_1}^{T_i'+t_1} \log(w_i - s_{it})e^{-\rho(t-t_1)}dt + \int_{T_i'+t_1}^{+\infty} [\log(w_i) + \beta]e^{-\rho(t-t_1)}dt$$
$$+ \lambda\left[\int_{t_{0i}}^{t_1} \bar{s}_{it}e^{-\pi_0(t_1-t_{0i})-\pi_1 T_i'}dt + \int_{t_1}^{T_i'+t_1} s_{it}e^{-\pi_1(T_i'+t_1)}dt - p_I\right]$$

The first-order conditions are:

$$T_i': \quad e^{-\rho T_i'}\{\log(w_i - s_{T_i'+t_1}) - [\log(w_i) + \beta]\} + \lambda[s_{T_i'+t_1} - \pi_1 p_I] = 0$$
$$s_{it}: \quad -\frac{e^{-\rho(t-t_1)}}{w_i - s_{it}} + \lambda e^{-\pi_1(T_i'+t_1-t)} = 0, \quad \forall t \in [t_1, T_i' + t_1]$$

With the multiplier $\lambda = e^{-\rho T_i'}/(w_i - s_{T_i'+t_1})$, the conditions become:

$$\log(w_i - s_{i,T_i'+t_1}) - [\log(w_i) + \beta] = \frac{\pi_1 p_I - s_{i,T_i'+t_1}}{w_i - s_{i,T_i+t_1}}$$
$$s_{i,T_i'+t_1} = w_i - \frac{\rho}{e^{\rho T_i'} - 1}\left[\frac{w_i}{\pi_1}(1 - e^{-\pi_1 T_i'}) - p_I\right]$$

From these two equations, T_i' and $s_{i,T_i'+t_1}$ can be found.

Solution of the model with temporary shock for extensive margin households with $T_i > t_2$

The Lagrangian is:

$$L = \int_{t_1}^{T_i'+t_1} u(w_i - s_{it}) e^{-\rho(t-t_1)} dt + \int_{T_i'+t_1}^{+\infty} [u(w_i) + \beta] e^{-\rho(t-t_1)} dt$$
$$+ \lambda \left[\int_{t_1}^{t_2} s_{it} e^{-\pi_1(t_2-t) - \pi_2(T_i'+t_1-t_2)} dt + \int_{t_2}^{T_i'+t_1} s_{it} e^{-\pi_2(T_i'+t_1-t)} dt - p_I \right]$$

The first-order conditions yield:

$$\begin{aligned}
T_i': \quad & \log(w_i - s_{T_i'+t_1}) - [\log(w_i) + \beta] = -\frac{s_{T_i'+t_1} - \pi_1 p_I}{w_i - s_{i,T_i'+t_1}} \\
s_{it}: \quad & s_{it} = w_i - (w_i - s_{i,T_i'+t_1}) e^{\rho(T_i'+t_1-t) + \pi_1(t_2-t) + \pi_2(T_i'+t_1-t_2)}, \quad \forall t \in [t_1, t_2] \\
& s_{it} = w_i - (w_i - s_{i,T_i'+t_1}) e^{(\rho+\pi_2)(T_i'+t_1-t)}, \quad \forall t \in [t_2, T_i' + t_1]
\end{aligned}$$

From these conditions and the constraint, T_i', $s_{i,T_i'+t_1}$, and the entire consumption and saving trajectory can be found.

Solution of the model with temporary shock for intensive margin households with $T_i > t_2$ born before t_1

The problem for these households is:

$$p_I = \int_{t_{0i}}^{t_1} \bar{s}_{it} e^{-\pi_0(t_1-t) - \pi_1(t_2-t_1) - \pi_2(T_i'+t_1-t_2)} dt + \int_{t_1}^{t_2} s_{it} e^{-\pi_1(t_2-t) - \pi_2(T_i'+t_1-t_2)} dt$$
$$+ \int_{t_2}^{T_i'+t_1} s_{it} e^{-\pi_2(T_i'+t_1-t)} dt$$

The first-order conditions are analogous to the previous case:

$$\begin{aligned}
T_i': \quad & \log(w_i - s_{T_i'+t_1}) - [\log(w_i) + \beta] = -\frac{s_{T_i'+t_1} - \pi_1 p_I}{w_i - s_{i,T_i'+t_1}} \\
s_{it}: \quad & s_{it} = w_i - (w_i - s_{i,T_i'+t_1}) e^{\rho(T_i'+t_1-t) + \pi_1(t_2-t) + \pi_2(T_i'+t_1-t_2)}, \quad \forall t \in [t_1, t_2] \\
& s_{it} = w_i - (w_i - s_{i,T_i'+t_1}) e^{(\rho+\pi_2)(T_i'+t_1-t)}, \quad \forall t \in [t_2, T_i' + t_1]
\end{aligned}$$

From these conditions and the constraint, T_i', $s_{i,T_i'+t_1}$, and the entire trajectory can be found.

Linear Probability Model

Table 18 describes the variables used in the Linear Probability Model.

Table 18: Description of variables

Variable	Description
*ad*85	1 if the household is in the 1985 PNAD or later
*ad*86	1 if the household is in the 1986 PNAD or later
*ad*87	1 if the household is in the 1987 PNAD or later
*ad*88	1 if the household is in the 1988 PNAD
luz	1 if the household has electricity
renda_domr	household income divided by average household income in that year
renda_domr2	renda_domr squared
tot_pes	total persons in the household
criancas1	1 if the household has 1 child
criancas2	1 if the household has 2 children
criancas3	1 if the household has 3 or more children
mulher	1 if the household head is a woman
mulher1	1 if the household has 1 woman
mulher2	1 if the household has 2 women
mulher3	1 if the household has 3 or more women
idade	age of the household head
idade2	age squared
anos_est	years of schooling of the household head
rural	1 if rural area
metrop	1 if metropolitan area
estados	dummies for each state

Table 19 presents the LPM results for the full sample (Total column) and for separate regressions for each quintile. Standard deviations are shown in parentheses. Asterisks indicate significance of the estimated coefficient.

Table 19: LPM for refrigerators, PNADs 1984–88 (omits state dummies)

Variables	Total	5th Quintile	4th Quintile	3rd Quintile	2nd Quintile	1st Quintile
*ad*85	−0.0159***	−0.00461	−0.000303	−0.00639	−0.00867***	−0.00342**
	(0.00161)	(0.00373)	(0.00435)	(0.00394)	(0.00290)	(0.00140)
*ad*86	−0.000525	0.00795*	0.0157***	0.00215	−0.0110***	−0.00314*
	(0.00179)	(0.00417)	(0.00476)	(0.00422)	(0.00328)	(0.00167)
*ad*87	0.000879	0.00336	0.00731	0.0051	0.00761**	0.00226
	(0.00191)	(0.00447)	(0.00505)	(0.00445)	(0.00342)	(0.00172)
*ad*88	0.00530***	0.0207***	0.0203***	0.0167***	0.00947***	−0.00094
	(0.00189)	(0.00441)	(0.00506)	(0.00451)	(0.00327)	(0.00160)
luz	0.516***	0.274***	0.442***	0.590***	0.705***	0.750***
	(0.00177)	(0.00293)	(0.00353)	(0.00415)	(0.00551)	(0.01080)
renda_domr	0.0208***	0.0855	0.916***	1.045***	0.441***	0.00120***
	(0.00202)	(0.11400)	(0.31300)	(0.19500)	(0.06260)	(0.00017)
renda_domr2	−0.000275**	1.649***	−0.383	−0.629***	−0.162***	−1.18e-05***
	(0.00012)	(0.43300)	(0.50500)	(0.17700)	(0.03100)	(0.00000)
tot_pes	0.00769***	0.00362***	−0.0127***	−0.0251***	−0.0194***	−0.00404***
	(0.00039)	(0.00095)	(0.00099)	(0.00091)	(0.00072)	(0.00044)
criancas1	0.00487***	−0.00674*	0.0282***	0.0460***	0.0305***	0.00818***
	(0.00149)	(0.00388)	(0.00415)	(0.00357)	(0.00253)	(0.00129)
criancas2	−0.00917***	−0.00792*	0.0328***	0.0616***	0.0415***	0.0101***
	(0.00180)	(0.00470)	(0.00494)	(0.00426)	(0.00312)	(0.00157)
criancas3	−0.0795***	−0.0456***	0.00501	0.0473***	0.0155***	−0.000925
	(0.00239)	(0.00561)	(0.00601)	(0.00553)	(0.00451)	(0.00271)
mulher	−0.0500***	0.00249	−0.00548	−0.0303***	−0.0244***	−0.00471***
	(0.00163)	(0.00329)	(0.00395)	(0.00364)	(0.00281)	(0.00147)
mulher1	0.0931***	0.187***	0.127***	0.191***	0.245***	0.226***
	(0.00416)	(0.00604)	(0.00951)	(0.01160)	(0.01200)	(0.00816)
mulher2	0.197***	0.110***	0.184***	0.250***	0.234***	0.0960***
	(0.00436)	(0.00701)	(0.01020)	(0.01200)	(0.01220)	(0.00824)
mulher3	0.216***	0.143***	0.205***	0.268***	0.249***	0.0998***
	(0.00433)	(0.00671)	(0.01000)	(0.01190)	(0.01210)	(0.00823)
idade	0.0166***	0.0128***	0.0203***	0.0199***	0.0109***	0.00277***
	(0.00025)	(0.00043)	(0.00057)	(0.00059)	(0.00049)	(0.00029)
idade2	−0.000145***	−0.000101***	−0.000163***	−0.000168***	−9.15e-05***	−2.30e-05***
	(0.00000)	(0.00000)	(0.00001)	(0.00001)	(0.00000)	(0.00000)
anos_est	0.0260***	0.0329***	0.0327***	0.0243***	0.0116***	0.00297***
	(0.00026)	(0.00066)	(0.00056)	(0.00042)	(0.00028)	(0.00013)
rural	−0.0559***	−0.0115***	−0.0251***	−0.0391***	−0.0452***	−0.0389***
	(0.00187)	(0.00312)	(0.00395)	(0.00429)	(0.00427)	(0.00437)
metrop	0.0594***	0.0874***	0.0653***	0.0394***	0.0173***	0.00888***
	(0.00144)	(0.00413)	(0.00391)	(0.00330)	(0.00242)	(0.00133)
Constant	−0.705***	−0.606***	−1.032***	−1.081***	−0.618***	0.0394**
	(0.00799)	(0.01380)	(0.05050)	(0.05640)	(0.03650)	(0.01550)
Observations	432392	86573	85087	88373	86215	86144
R^2	0.472	0.314	0.338	0.342	0.339	0.36

* significant at 10%; ** significant at 5%; *** significant at 1%.

Computational Strategy

Constant inflation The parameter values to be used in the simulation are first defined: $\{\beta, \rho, \pi_0, p_I, w_\mu, w_{\sigma^2}\}$, where the last two refer to the income distribution to be generated from a log-normal; additionally, N (the number of new households each period) and T_{final} (the number of periods to be simulated) are set.

A loop is then run with households choosing the optimal consumption, saving, and purchase time trajectories for each one, through the model's first-order conditions (FOCs). After this calculation, the household's utility under the FOC trajectory is compared to the utility of always consuming the non-durable good. If utility is higher when the household always consumes non-durables, then it will not purchase the durable good; its saving will always be zero and its non-durable consumption will always equal its income.

Permanent Shock In this case, the simulation computes all initial parameters identically to the constant inflation case, plus two additional parameters: t_1 and π_1, where t_1 is the period in which inflation transitions from π_0 to π_1. At this point, households that were saving (intensive margin) re-evaluate their consumption trajectory and solve the new problem. Additionally, households that were not purchasing re-evaluate their decision. The FOCs are used to find the consumption, saving, and purchase time trajectories. Again, the household's utility under the FOC trajectory is compared to that of always consuming the non-durable. If the latter is higher, the household will not purchase the durable good; otherwise, it becomes part of the extensive margin.

Transitory Shock In this case, the simulation computes all initial parameters identically to the constant inflation case, plus four additional parameters: t_1, t_2, π_1, π_2, where t_1 is the period in which inflation transitions from π_0 to π_1, and t_2 is the period in which inflation transitions from π_1 to π_2. At this point, households that were saving (intensive margin) re-evaluate their consumption trajectory and solve the new problem. Additionally, households that were not purchasing re-evaluate their decision. The FOCs are used to find the trajectories. Again, the household's utility is compared to that of always consuming the non-durable. The quasi-intensive margin represents households born during the transition period that evaluate their optimal trajectories from the FOCs and only purchase after period t_2.